CS430/910: Foundations of Data Analytics

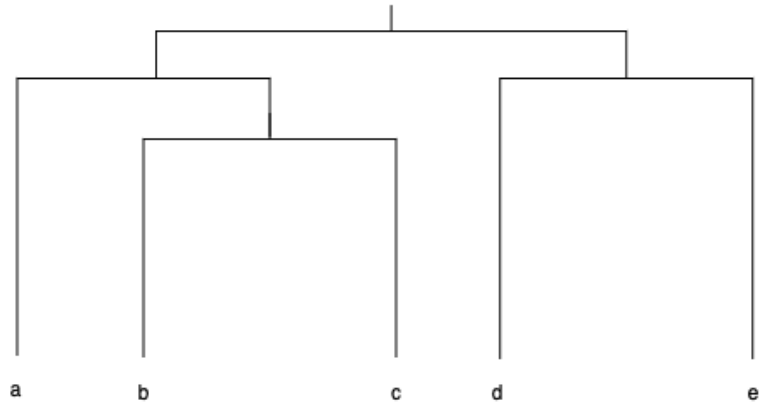
Clustering 2 | Dr Ayse Sunar

Objectives

- To understand the concept of clustering and clustering methods
- To see different clustering methods and objectives:
  - k**-means**
  - k**-medoids**
  - **Hierarchical Agglomerative Clustering**
  - **DBSCAN**
- To use tools (Weka) to cluster data and study the results

Hierarchical Clustering

- We may wish for data to be organised hierarchically:
  - For example, may want a cluster to represent all modules at the university. You could then partition the cluster into smaller clusters, where each cluster represents modules from an individual department.
    - Could then divide clusters even further, based on year of study!
- Useful for data visualisation!
  - Can view hierarchy of data as a tree.



Hierarchical Clustering

- Hierarchical clustering creates a hierarchical decomposition of a set of data objects.
- Two different types:
  - **Agglomerative Approach** (also called a **Bottom-Up Approach**): Starts with each object in a separate cluster, and continuously merges clusters until a terminating condition is met.
  - **Divisive Approach** (also called a **Top-Down Approach**): Starts with all objects in the same cluster, and divides the clusters until each object is in its own cluster, or a terminating condition is met.

Hierarchical Clustering

Agglomerative Clustering

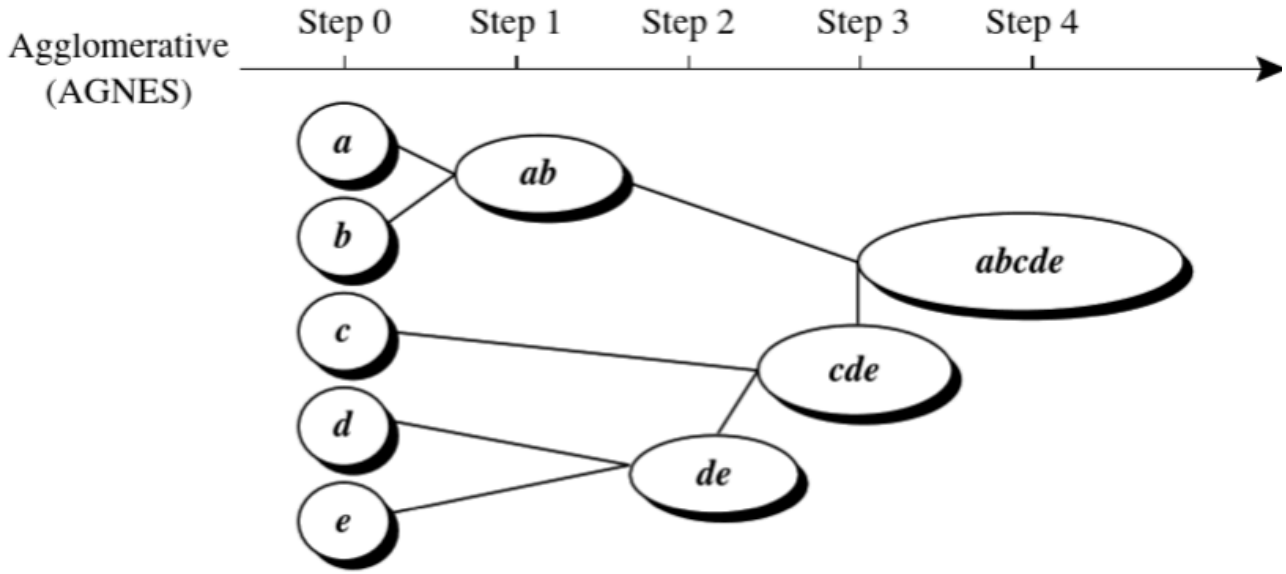
- **Agglomerative Approach** (also called a **Bottom-Up Approach**): Starts with each object in a separate cluster, and continuously merges clusters until a terminating condition is met.
- **Merging Step**: Find the two clusters that are the closest to each other (by some similarity measure), and merge into one cluster.

Hierarchical Clustering

Agglomerative Clustering - Example

- AGNES (***AG****lomerative* ***NES****ting*) is an example of an agglomerative hierarchical clustering approach.

- Let X be a set of five data objects, $\{a, b, c, d, e\}$.



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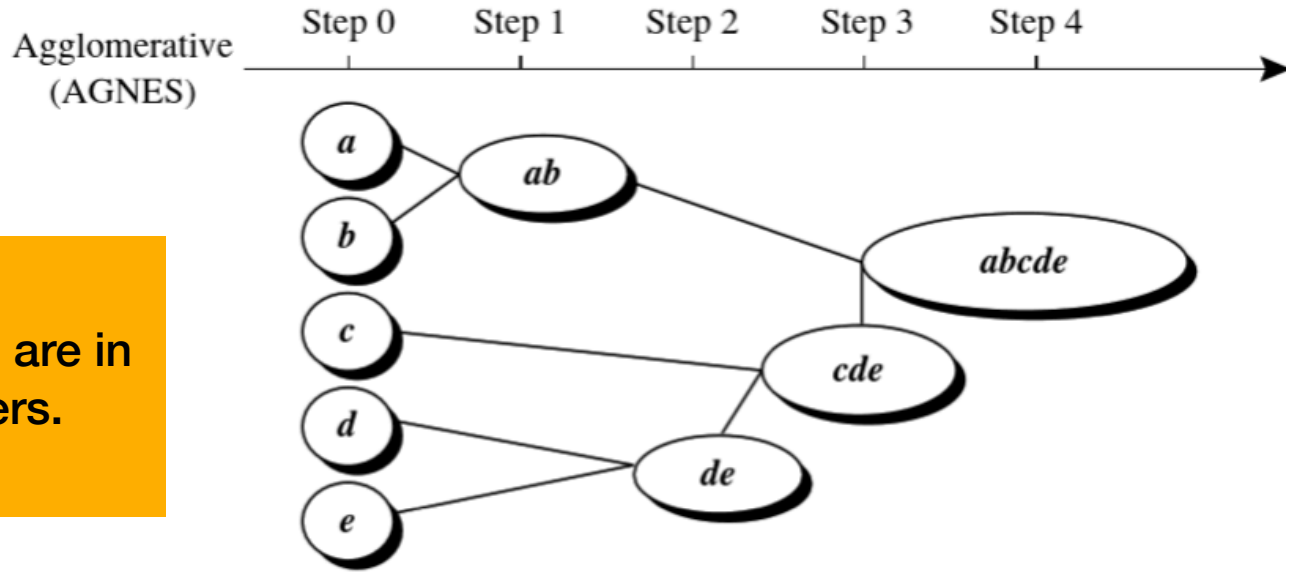

Step 0: All objects are in separate clusters.

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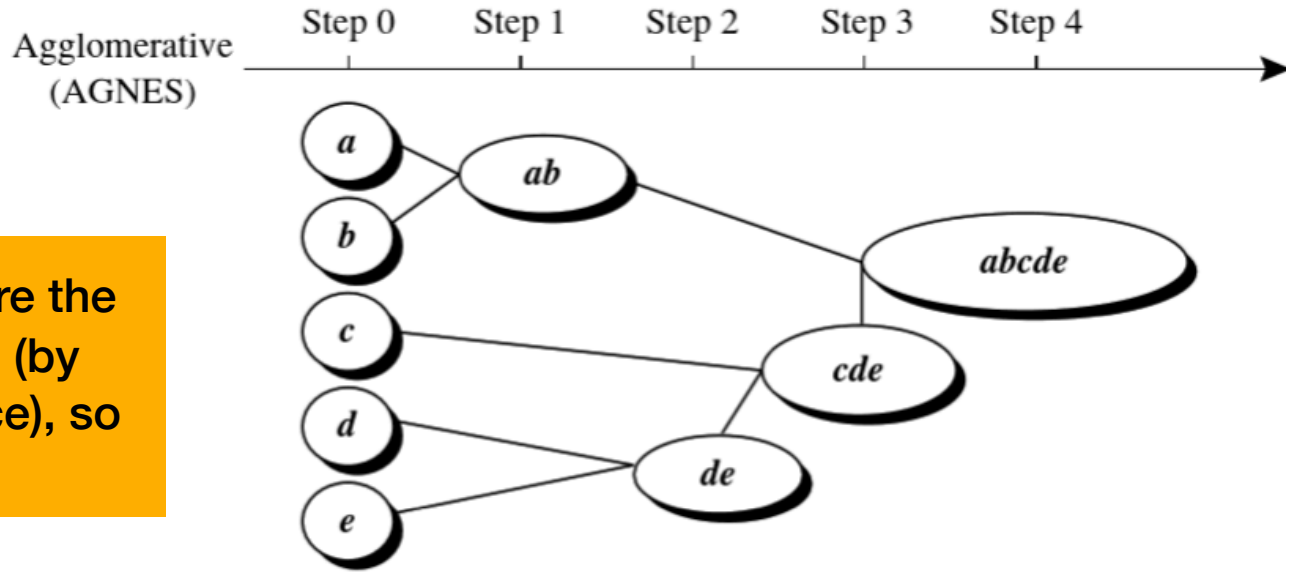

Step 1: a and b are the closest clusters (by Euclidean distance), so merge.

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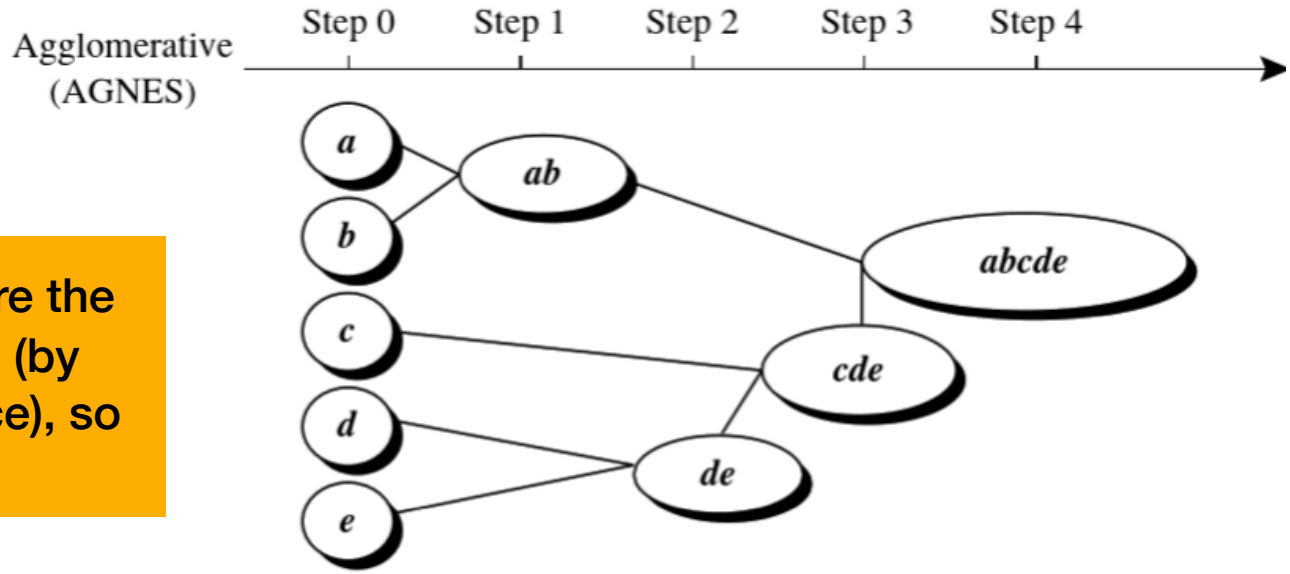

Step 2: d and e are the closest clusters (by Euclidean distance), so merge.

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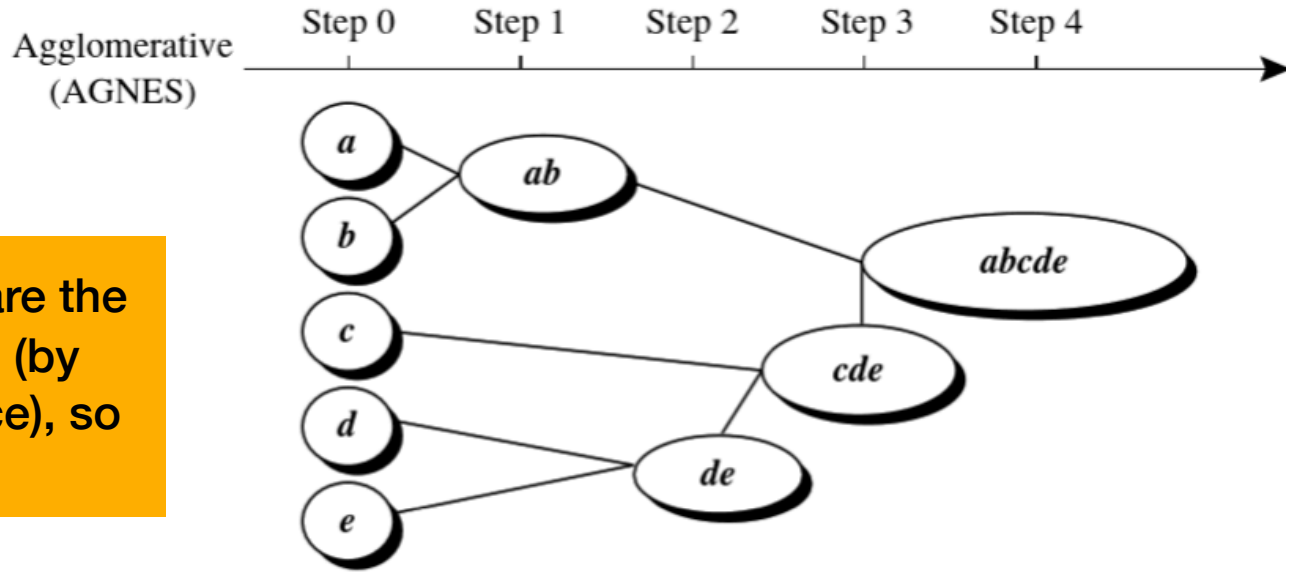

Step 3: c and de are the closest clusters (by Euclidean distance), so merge.

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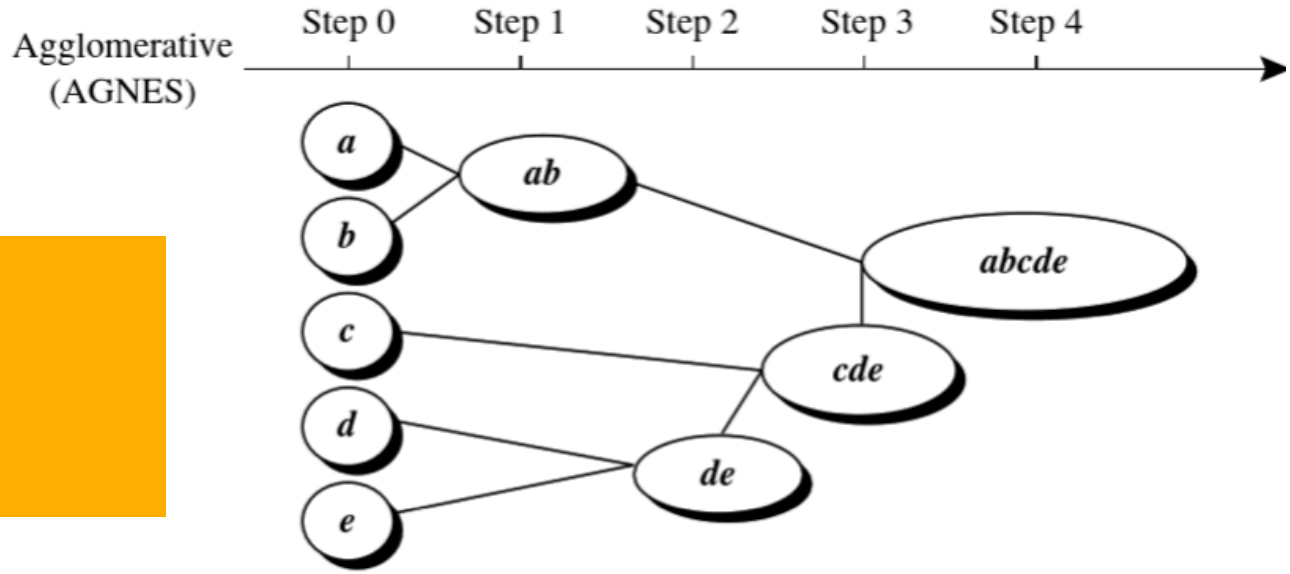

Etc.

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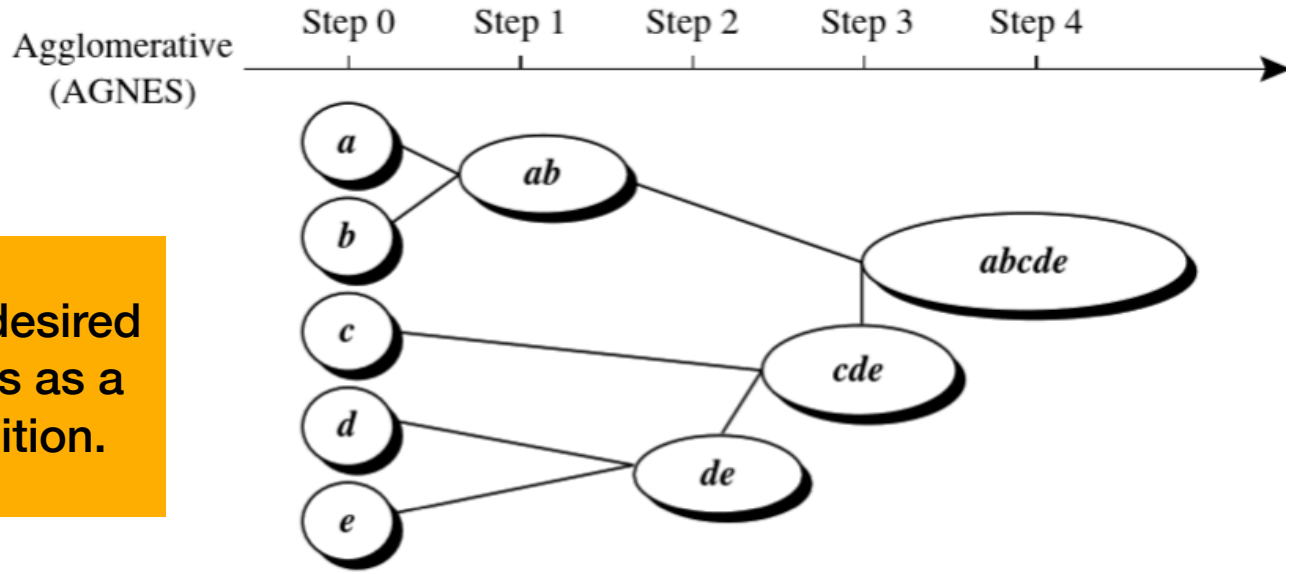

We can specify a desired number of clusters as a terminating condition.

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Hierarchical Clustering

Divisive Clustering

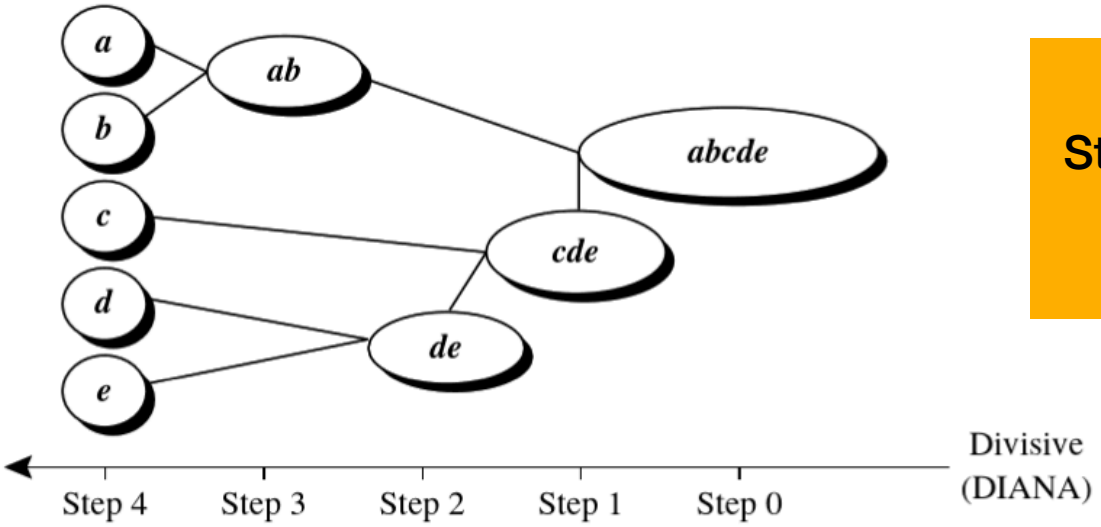
- **Divisive Approach** (also called a **Top-Down Approach**): Starts with all objects in the same cluster, and divides the clusters until each object is in its own cluster, or a terminating condition is met.
- **Splitting Step**: Split according to some principle, example in next few slides.

Hierarchical Clustering

Divisive Clustering - Example

- DIANA (***DI****visive* ***ANA****lysis*) is an example of a divisive hierarchical clustering approach.

- Let X be a set of five data objects, $\{a, b, c, d, e\}$.



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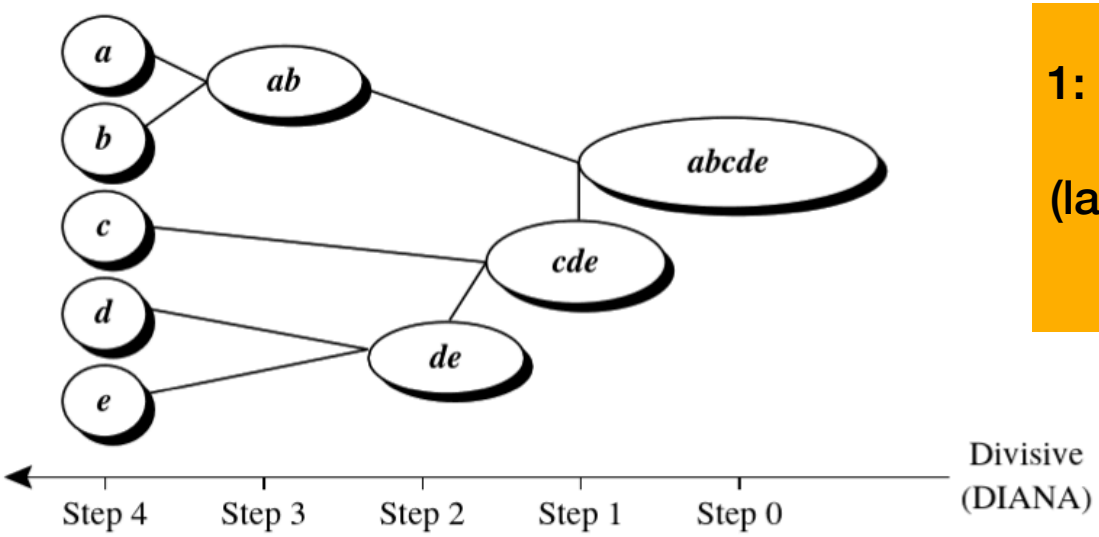

1: Choose the cluster with the largest *diameter* (largest distance between any two objects)

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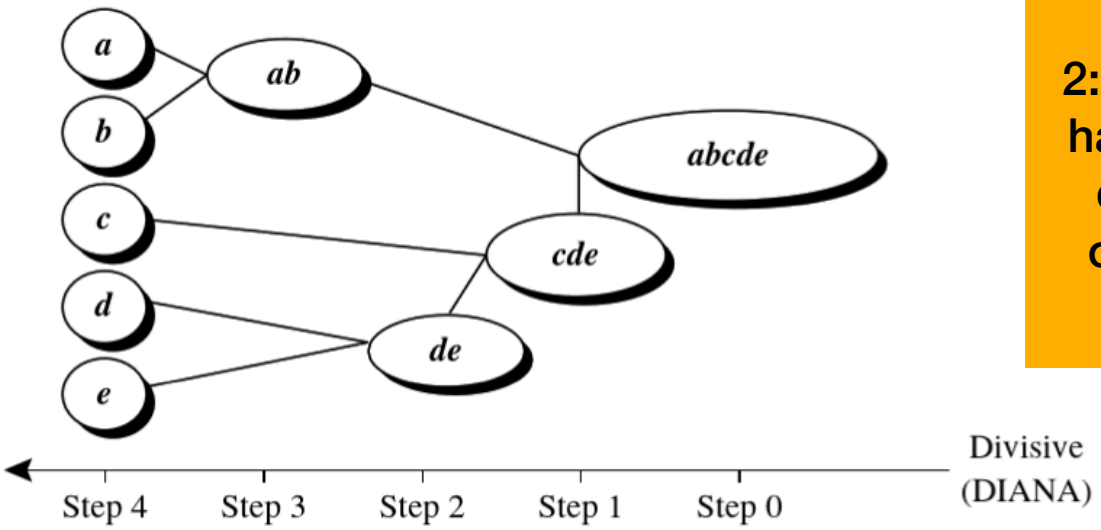

2: Find the object which has the largest average distance to the other objects of the cluster.

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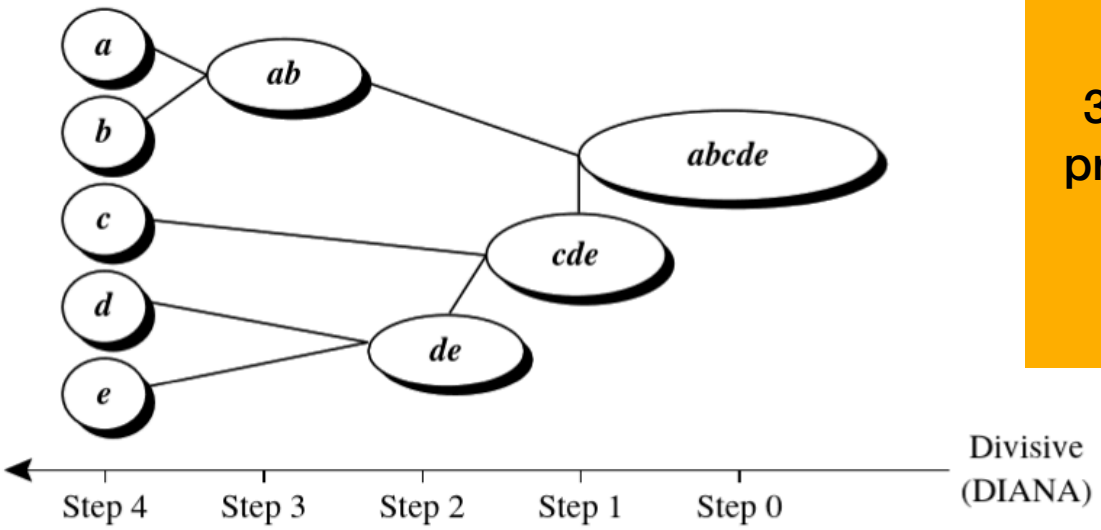

3: The object from the previous step initiates a *splinter group*.

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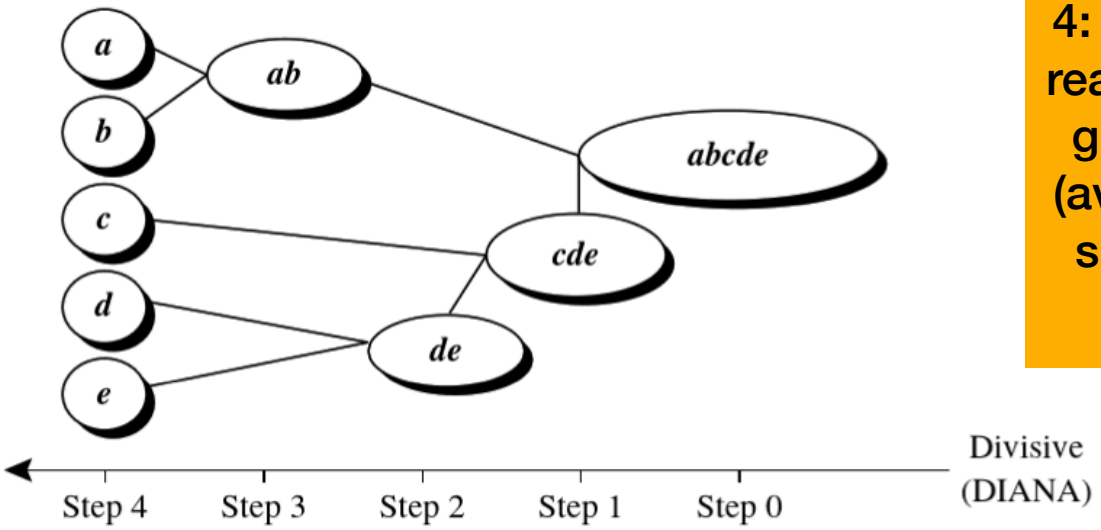

4: Remaining objects are reassigned to the splinter group if they are closer (average distance) to the splinter group than the original cluster.

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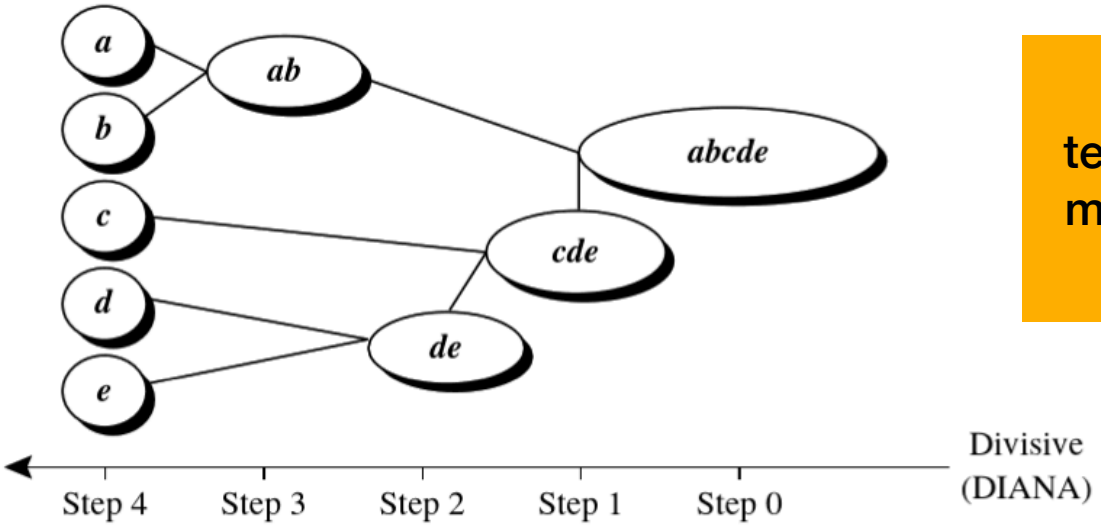

Repeat 1 to 4 until terminating condition is met or all objects are in separate clusters.

Hierarchical Clustering

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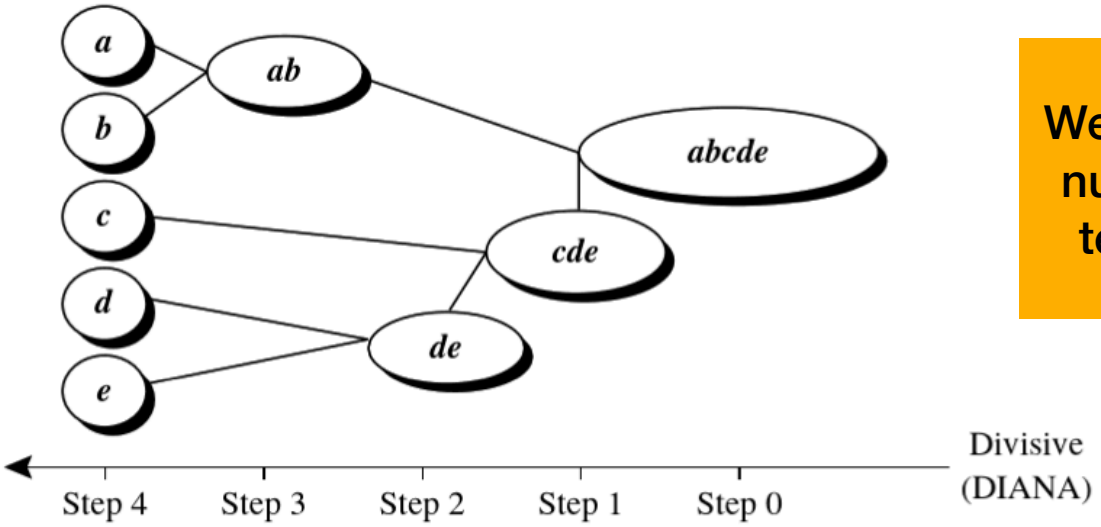
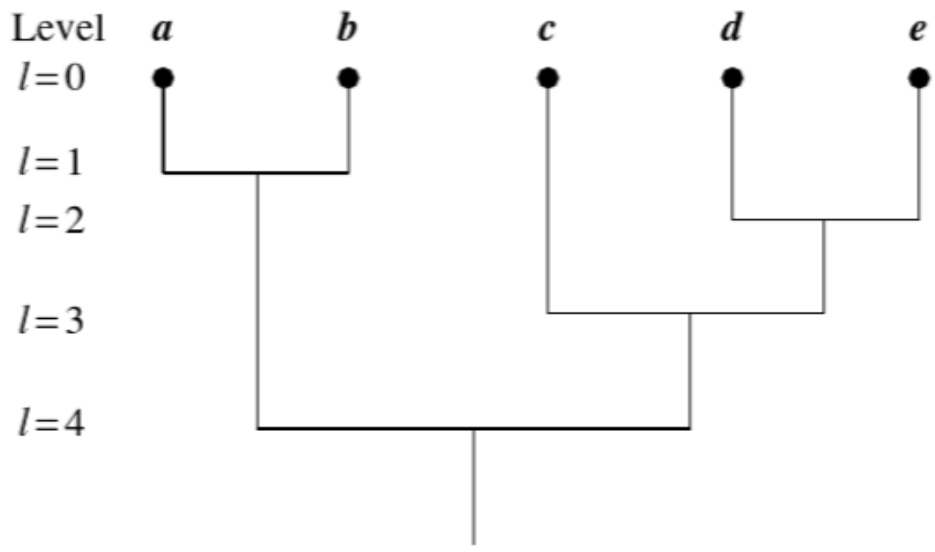


We can specify a desired number of clusters as a terminating condition.

Dendrograms

- A ***Dendrogram*** is a tree-like structure that shows how objects are grouped together (agglomerative clustering) or partitioned (divisive clustering).
- Very common for visualisation of clusters.
- Can cut off at certain levels to choose a certain number of clusters (e.g. splitting at $l = 2$ would leave three clusters (ab, c, de)).



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Peer Discussion (2 mins)

- Is it similar to Decision Trees?
- How similar?
- How different?

Peer Discussion (2 mins)

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- How similar?
- How different?

vevox

Join at:

vevox.app

ID:

131-565-585

Distance Measures

- How can we measure the distance between clusters to find the closest pair?

  1. **Single-Link**: $d(C_1, C_2) = min\ D(c_1 \in C_1, c_2 \in C_2)$
  2. **Complete-Link**: $d(C_1, C_2) = max\ D(c_1 \in C_1, c_2 \in C_2)$
  3. **Average-Link**: $d(C_1, C_2) = avg\ D(c_1 \in C_1, c_2 \in C_2)$

  where $d(C_1, C_2)$ is the distance between clusters C_1 and C_2, and $D(c_1 \in C_1, c_2 \in C_2)$ is the distance between two objects, $c_1 \in C_1$ and $c_2 \in C_2$.

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3. **Average-Link**: $d(C_1, C_2) = avg\ D(c_1 \in C_1, c_2 \in C$

Distance is determined by a single pair of elements (one in each cluster).

where $d(C_1, C_2)$ is the distance between clusters C_1 and
$D(c_1 \in C_1, c_2 \in C_2)$ is the distance between two objects, $c_1 \in C_1$ and $c_2 \in C_2$.

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3. **Average-Link**: $d(C_1, C_2) = avg\ D(c_1 \in C_1, c_2 \in C_2)$

Distance is determined by the average of all distances from pairs of elements (one in each cluster).

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Hierarchical Clustering

Cost (Agglomerative Example)

- Hierarchical Clustering can be costly to implement:
  - Initially, must compute distance from every object to every other object.
  - Each merge requires a new computation of distances involving the merged clusters.
- **This limits scalability**: with only few hundred thousand points, the clustering could take days or months.
  - Need clustering methods that can scale to allow processing of large data sets.

Single Link Clustering in Weka

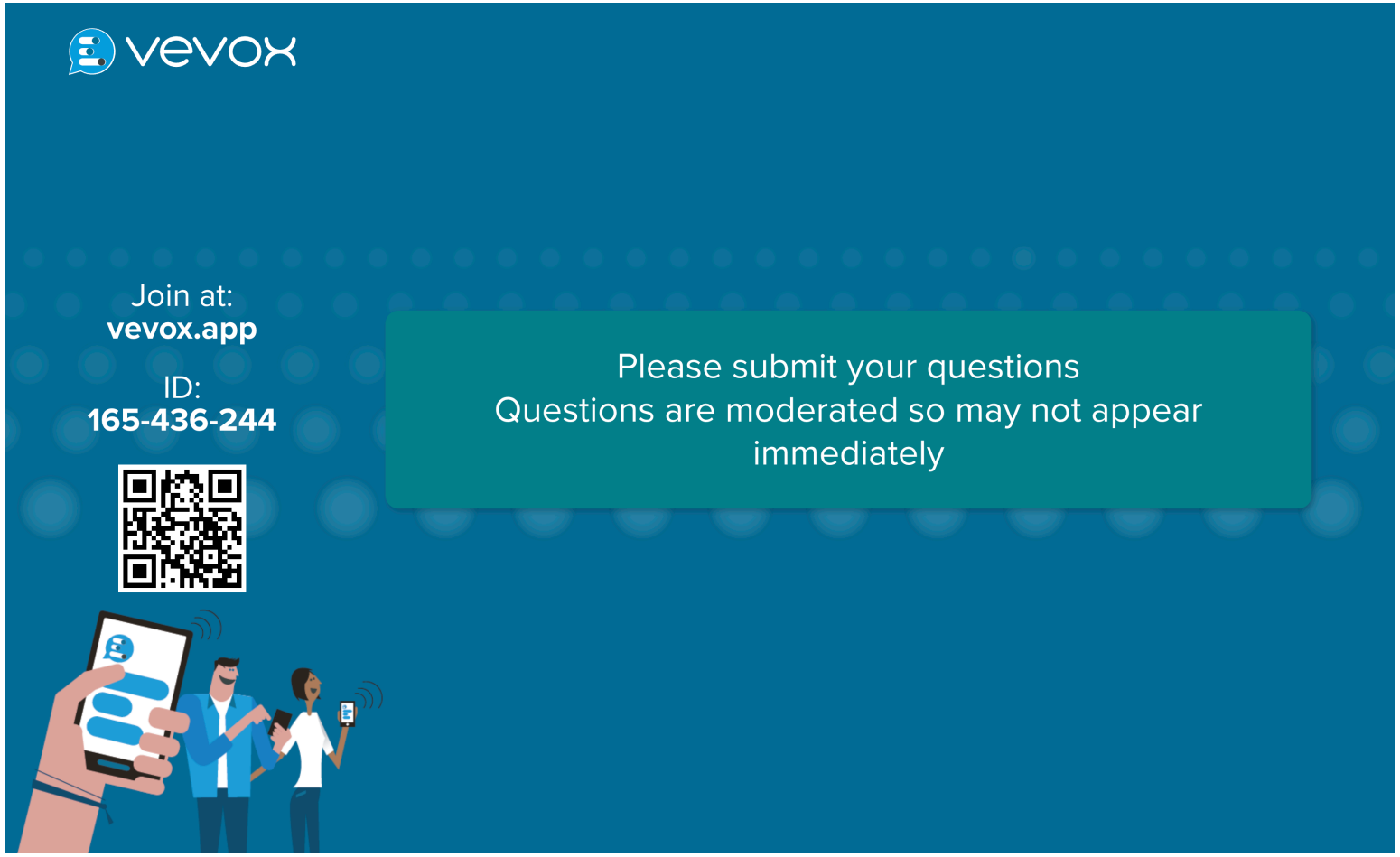
```
=== Run information ===
Scheme:weka.clusterers.HierarchicalClusterer -N 3 -L SINGLE -P -A
    "weka.core.EuclideanDistance -R first-last"
Relation:     iris
=== Model and evaluation on training set ===
Time taken to build model (full training data) : 0.17 seconds
Clustered Instances
0       49 ( 33%)
1        1 (  1%)
2      100 ( 67%)

Classes to Clusters:
  0  1  2  <-- assigned to cluster
 49  1  0 | Iris-setosa
  0  0 50 | Iris-versicolor
  0  0 50 | Iris-virginica
Cluster 0 <-- Iris-setosa
Cluster 1 <-- No class
Cluster 2 <-- Iris-versicolor

Incorrectly clustered instances : 51.0  34      %
```

Average Link Clustering in Weka

```
Scheme:weka.clusterers.HierarchicalClusterer -N 3 -L AVERAGE -P -A
    "weka.core.EuclideanDistance -R first-last"
Relation:     iris
Test mode:Classes to clusters evaluation on training data
=== Model and evaluation on training set ===
Time taken to build model (full training data) : 0.08 seconds

Clustered Instances
0       50 ( 33%)
1       67 ( 45%)
2       33 ( 22%)

Classes to Clusters:
  0  1  2  <-- assigned to cluster
 50  0  0 | Iris-setosa
  0 50  0 | Iris-versicolor
  0 17 33 | Iris-virginica

Cluster 0 <-- Iris-setosa
Cluster 1 <-- Iris-versicolor
Cluster 2 <-- Iris-virginica
Incorrectly clustered instances : 17.0  11.3333 %
```

Quiz Time

Revising our knowledge on regression, classification and clustering



Acknowledgments

- Greg Watson [Warwick, CS430/CS910]
- Han, J., Pei, J. and Kamber, M., 2012. *Data Mining: Concepts and Techniques*. Elsevier.
- Graham Cormode [Warwick, CS910]
- Florin Ciucu [Warwick, CS430/CS910]